

Foundation (Jalapeno)

- Q1.** A basketball player scores x points in a game, and her average score for the season is $x - 2$. If she has played 5 games, what system of equations represents this situation?
- (A) $x + 5 = 25$, $x - 2 = 5$
 (B) $5x = 25$, $x - 2 = 5$
 (C) $x + 2 = 5$, $5x = 25$
 (D) $5x = 25$, $x = 5$
 (E) $x - 2 = 5$, $5x = 25$
- Q2.** The number of fans y at a soccer game is related to the number of goals x scored by the home team. The relationship can be modeled by $y = 2x + 100$ and $y = x^2 - 4x + 50$. What is the point of intersection between the two graphs?
- (A) (0, 100)
 (B) (2, 100)
 (C) (4, 50)
 (D) (6, 80)
 (E) (8, 100)
- Q3.** A cyclist travels x miles in y hours. Her speed can be represented by $y = x - 2$ and $y = x^2 - 4x + 3$. What is the cyclist's speed when $x = 4$?
- (A) 0 mph
 (B) 1 mph
 (C) 2 mph
 (D) 3 mph
 (E) 4 mph
- Q4.** A sports equipment company produces x units of a product each day. The cost y of producing x units is given by $y = 2x + 50$ and $y = x^2 + 20x - 100$. What is the minimum number of units that must be produced to break even?
- (A) 0 units
 (B) 5 units
 (C) 10 units
 (D) 15 units
 (E) 20 units
- Q5.** A swimmer's time y in seconds to complete x laps can be represented by $y = 2x + 10$ and $y = x^2 + 4x$. What is the point of intersection between the two graphs?
- (A) (0, 10)
 (B) (2, 14)
 (C) (-2, 0)
 (D) (4, 22)
 (E) (-4, -2)
- Q6.** A runner's heart rate y in beats per minute is related to the distance x she runs in miles. The relationship can be modeled by $y = x + 60$ and $y = x^2 - 2x + 40$. What is the point of intersection between the two graphs?
- (A) (0, 60)
 (B) (2, 40)
 (C) (4, 60)
 (D) (6, 70)
 (E) (8, 80)
- Q7.** A golfer's score y is related to the number of strokes x she takes. The relationship can be modeled by $y = 2x - 10$ and $y = x^2 - 4x$. What is the point of intersection between the two graphs?
- (A) (0, -10)
 (B) (2, -6)
 (C) (4, 0)
 (D) (6, 10)
 (E) (8, 20)
- Q8.** A tennis player's serve speed y in miles per hour is related to the force x she applies. The relationship can be modeled by $y = x + 20$ and $y = x^2 - 4x + 10$. What is the point of intersection between the two graphs?
- (A) (0, 20)
 (B) (2, 10)
 (C) (4, 20)
 (D) (6, 30)
 (E) (8, 40)

Open Ended Questions (FRQ)

- Q9.** A cyclist travels x miles in y hours. Her speed can be represented by $y = x - 2$ and $y = x^2 - 4x + 3$. Find the point of intersection between the two graphs and interpret its meaning in the context of the situation. Show all work and explain your answer.

- Q10.** A sports team's winning percentage y is related to the number of games x they win. The relationship can be modeled by $y = 0.2x + 0.4$ and $y = 0.01x^2 + 0.1x + 0.2$. Find the point of intersection between the two graphs and interpret its meaning in the context of the situation. Show all work and explain your answer.

Chef's Tips

- Tip 1: Remind students to read each question carefully and identify the key elements of the problem.
- Tip 2: Encourage students to use graphing tools or algebraic methods to find the point of intersection between the two graphs.
- Tip 3: Emphasize the importance of interpreting the results in the context of the situation and explaining their answer clearly.